HUTECH INSTITUTE OF INTERNATIONAL EDUCATION

RESEARCH METHODOLOGY

TOPIC: THE IMPACT OF SOCIAL MEDIA ON STUDENTS' ACADEMIC PERFORMANCE

Subject: RESEARCH METHODOLOGY

Specialty: DIGITAL MARKETING

Supervisor (Lecturer): NGUYỄN VĂN HẬU

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3. **Student's name: PHAN NGỌC THANH HỒNG Student's ID:2254040014**

Ho Chi Minh City, 2025.

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Although we have tried, due to limited knowledge, our scientific research topic will certainly not be able to avoid many errors. We look forward to receiving additional comments and amendments from you so that our research paper can be more complete.

We sincerely thank you!

INSTRUCTOR COMMENTS

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Instructor

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LIST OF ABBREVIATIONS

TTF - Task Technology Fit

TAM 3 - Technology Acceptance Model 3

SMAS - Social Media Addiction Scale

WHO 5 - World Health Organization - 5

UWES-S - Utrecht Work Engagement Scale – Student version

DigComp (EU) - Digital Competence Framework (European Union)

GPA - Grade Point Average

EFA - Exploratory Factor Analysis

CFA - Confirmatory Factor Analysis

AMOS - Analysis of Moment Structures

VIF - Variance Inflation Factor

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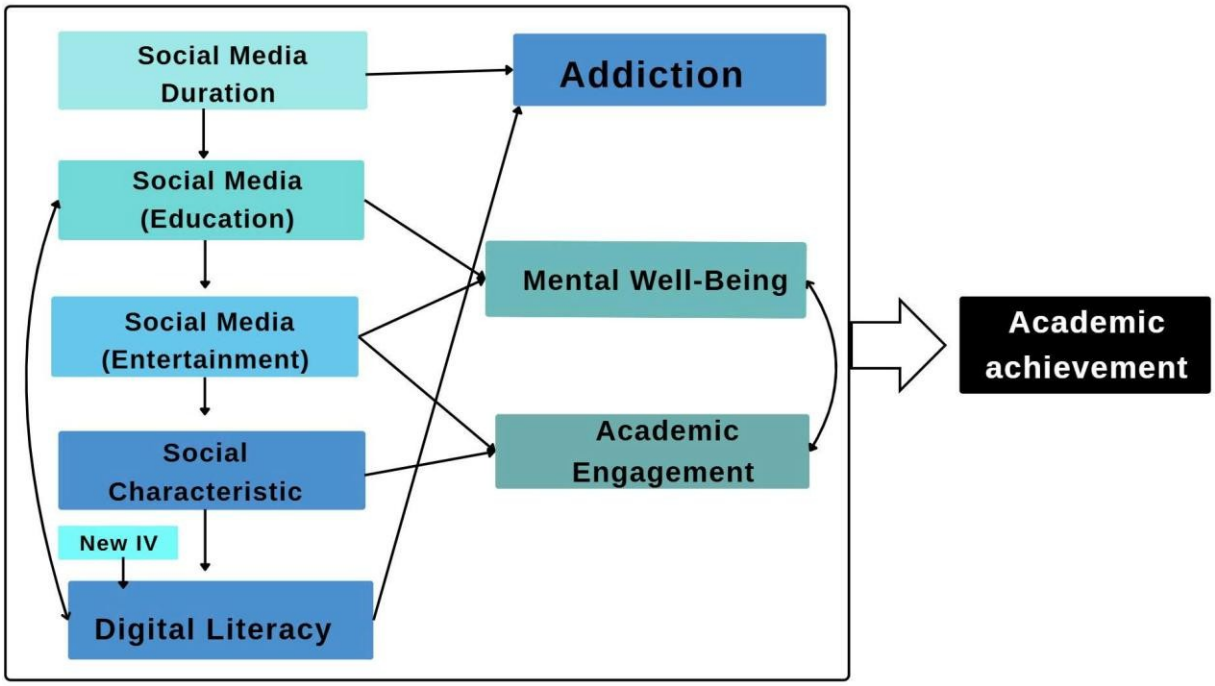


Figure 1: Theoretical model.

ASSIGNMENT 1

1. Title of the Study

The impact of social networks on students' academic performance

2. Introduction

With the development of information technology and the strong development of the internet everywhere, and it is a spiritual wave that shows no signs of stopping, it has become an indispensable part of everyone from young people to the elderly. It can be said that the amount of time people spend on social networks is more than the time spent with friends, family, studying, even in studying, the phenomenon that is most noticeable is the phenomenon that cannot be separated from social networks of students today.

Education is considered an important foundation for promoting sustainable development and long-term economic growth of each country. Ineffective learning leads to a shortage of quality and qualified human resources in developing countries. Therefore, improving learning performance has become a key task and also a major challenge for educational institutions. In recent years, many educational institutions have made efforts to improve learning outcomes as well as enhance the quality of teaching and learning. The core goal of the education system is still to achieve high learning outcomes, to create the best results while reducing the rate of low-achieving students (Phauk Sokkhey & Takeo Okazaki, 2020).

This has a significant impact on online learning outcomes as well as student GPAs. Research suggests that the use of technology, especially the internet, plays a pivotal role and can have a positive or negative impact on academic performance. When students spend too much time on social media platforms like Facebook, they are less likely to have the time to focus on their studies. As a result, their performance declines throughout their secondary school years, making it difficult to continue pursuing higher education (Saman Amjad et al., 2022).

Platforms such as Instagram, Facebook, YouTube, Telegram, and TikTok are now not only serving as entertainment tools, but also becoming informal learning channels with high flexibility and convenient accessibility. The integration of social media into the education sector, especially in the context of higher education, is increasingly being prioritized due to its ability to provide modern learning opportunities to students. These platforms make it easy for students to access learning content, while enhancing interaction with classmates and industry experts (Jamal Abdul Nasir Ansari & Nawab Ali Khan, 2020).

However, the rate of students using social media during class is also increasing, which affects concentration and the quality of learning is increasingly declining and affects the mood of teachers a lot, creating a great distance between students and students. Many studies have shown that integrating social media into educational technology tools in higher education has positive impacts, especially for students. However, it is important to be cautious when using these platforms in educational settings. While social media should not be completely eliminated, it is necessary to monitor and use it appropriately. When used properly, social media can contribute to improving student satisfaction and learning outcomes (Ali Mugahed Al-Rahmi et al. , 2020).

3. Background of the Study

In today's digital age, social media platforms have become an indispensable part of students' personal and academic lives, creating an impact in two trends - on the one hand, facilitating learning, on the other hand, causing distraction affecting the learning of Gen Z youth at that time. Due to the current development of technology, Gen Z has become too dependent on it, the strong development of platforms such as Facebook, WhatsApp and Instagram, tiktok... in the educational context has raised the question of when and how social media can act as a beneficial tool for academic achievement, and under what conditions it can become counterproductive (Ting Tin Tin et. al. , 2024).

The study of applied the Technology-Task Fit (TTF) model, arguing that the effectiveness of technology depends on the degree of fit between technology characteristics and learning tasks. In a university environment, when social media features such as accessibility, interactivity, and quick feedback are suitable for tasks such as knowledge sharing, group discussions, and academic exchanges, social media become effective learning support tools. The survey results of 206 students showed that technology characteristics, social characteristics, and task-technology fit significantly influenced the intention to use social media for learning, thereby improving learning performance and satisfaction. In the same vein, the study by Barabadi et. al., xtended the TAM 3 model to assess how students perceive social media in learning. In a sample of 400 agricultural students, the study found that factors such as usefulness, ease of use, quality of outcomes, social norms, and enjoyment all influence behavioral intentions, which in turn lead to actual social media use and improved learning outcomes. The combination of rational factors (such as perceived learning value) and emotional factors

(pleasure from use) helps to better explain how students sustain their use of social media for educational purposes.

In contrast to the above views, took a more cautious view by examining the psychosocial risks of social media use among Malaysian university students. Their study highlighted social media addiction among students. Despite the negative consequences, this factor did not clearly and accurately demonstrate its direct impact on students' academic performance according to the tested model. This has caused an inconsistency in the conclusion that social media can both enhance learning and cause negative problems and hinder learning if used in the wrong way or for the wrong purpose. This raises a major concern about excessive digital exposure affecting students' health and concentration, thereby indirectly affecting their ability to engage in effective learning. Although previous studies have identified a link between social media use and academic performance, there is still a significant gap in understanding the role of intrinsic motivational factors, student engagement in learning activities, and self-regulation of learning behavior. These factors may play a role in explaining why social media has increasingly influenced student learning, a point that is lacking in many current studies. These studies highlight an important insight: whether social media benefits or hinders academic performance - its impact is highly dependent on the context, shaped by the purpose, mode, and intensity of use. When integrated with understanding and used in a specific and appropriate way, it can promote more productive and effective learning, but if left unchecked, it can be distracting or psychologically stressful. Although previous studies such as the TTF and TAM 3 models have identified a link between social media use and academic performance, there is still a significant gap in understanding the role of motivational factors, student engagement in learning activities, and self-regulation of learning behavior. These factors may play a role in explaining why social media may or may not help learning (called the mediating role), or alter the impact of social media depending on the individual or situation (called the moderating role). This is a missing point in many current studies. Overall, exploring these mechanisms is necessary to gain a deeper understanding of when, for whom, and under what circumstances social media actually has a positive effect on learning (Ting Tin Tin et. al. , 2024).

4. Problem Statement

Digital technology, especially social media, has significantly changed students' learning behavior. In recent years, social media has become an indispensable part of students' daily lives. Platforms such as Facebook, TikTok, Instagram, or YouTube not only help students connect

and entertain but also support learning and update information etc. not only for entertainment but also for learning, leading to both benefits and challenges such as loss of concentration, procrastination and access to misinformation. Uncontrolled use of social media can negatively affect students' ability to concentrate and manage their study time. Although there are many general studies on the relationship between social media and learning, there is a lack of studies focusing on specific contexts (such as at ISB Atma Luhur in Pangkalpinang), and especially the lack of cases explaining how learning behaviors are transformed and directly impact students' academic performance in a local university environment (Ting Tin Tin et. al. , 2024).

Adding to the above arguments, a study by Khan and Subhan in Bangladesh showed a worrying link between social media addiction and poor academic performance. Accordingly, students tend to prioritize social media over studying, relaxing or real-life interactions. As a result, their grade point averages drop sharply, accompanied by symptoms such as lack of motivation, procrastination, and increased feelings of anxiety and stress. These findings suggest that social media dependence is not only a personal problem, but also a common challenge for the school environment in developing countries, including Vietnam. Therefore, further research in local contexts is needed to clearly identify the extent of influence and propose effective control solutions (Khan Abdus Subhan et. al. , 2025).

Examples of the use of social media among university students in Malaysia have become a common phenomenon and are diverse in form and purpose. Social media is not only a tool for entertainment and communication but also plays an important role in supporting learning and developing personal skills. However, besides the obvious benefits, the use of social media also has many potential negative impacts on students' learning outcomes and mental health. Here are some of the benefits of using social media through online study groups, knowledge exchange forums, students can easily discuss, share documents and solve academic problems together. This promotes collaborative learning, improves communication skills and teamwork, important skills in the modern university environment. However, the use of social media also poses many significant challenges. Social media addiction is also on the rise, making it difficult for students to manage their time and prioritize their academic tasks. In addition, cyberbullying has a serious impact on students' mental health, causing feelings of loneliness, anxiety, and depression. The line between using social media for academic and recreational purposes is becoming increasingly blurred, making it difficult for students to balance their academic responsibilities and social needs. The pressure to maintain a personal image on social

media, along with comparing oneself to others, can lead to psychological stress, anxiety, and feelings of insecurity. These issues not only affect mental health but also reduce motivation and active participation in learning activities. Furthermore, spending a lot of time on social media can reduce the time spent on physical activities and sleep, thereby negatively affecting students' overall health and academic performance. The blurred line between academic and social media use complicates students' ability to balance their responsibilities, potentially leading to anxiety, depression, and reduced academic engagement (Ting Tin Tin et. al. , 2024).

Despite the increasing popularity of social media in the digital age, its impact on academic performance remains understudied, especially in developing countries. In addition, the mediating role of smart learning environments in the relationship between social media use and academic performance remains largely unexplored, especially in Kenya. To fill these research gaps, the current study focuses on analyzing the extent to which social media influence academic performance of university students through the impact of smart learning environments. The study is based on four main hypotheses (Hamidah & Okkita Rizan, 2025).

5. Research Question

Lesson 1: Impact of Social Media on Undergraduate Students' Academic Performance in Malaysia

H1: Social media duration is positively related to social media addiction.

H2: Social media duration is positively related to cyberbullying.

H3: Social media use for education is positively related to social media addiction.

H4: Social media use for education is positively related to cyberbullying.

H5: Social media use for entertainment/self-pleasure is positively related to addiction.

H6: Social media use for entertainment/self-pleasure is positively related to cyberbullying.

H7: Social media addiction is positively related to mental well-being.

H8: Cyberbullying is positively related to mental well-being.

H9: Mental well-being is positively related to academic performance.

H10: Social media addiction mediates the relationship between social media duration and mental well-being.

H11: Cyberbullying mediates the relationship between social media duration and mental well-being.

H12: Social media addiction mediates the relationship between social media use (education) and mental well-being.

H13: Cyberbullying mediates the relationship between social media use (education) and mental well-being.

H14: Social media addiction mediates the relationship between social media use (entertainment/self-pleasure) and mental well-being.

H15: Cyberbullying mediates the relationship between social media use (entertainment/self-pleasure) and mental well-being.

H16: Mental well-being mediates the relationship between social media addiction and academic performance.

H17: Mental well-being mediates the relationship between cyberbullying and academic performance.

H18: Addiction and mental well-being jointly mediate the relationship between social media duration and academic performance.

H19: Cyberbullying and mental well-being jointly mediate the relationship between social media duration and academic performance.

H20: Addiction and mental well-being jointly mediate the relationship between social media use (education) and academic performance.

H21: Cyberbullying and mental well-being jointly mediate the relationship between social media use (education) and academic performance.

H22: Addiction and mental well-being jointly mediate the relationship between social media use (entertainment/self-pleasure) and academic performance.

H23: Cyberbullying and mental well-being jointly mediate the relationship between social media use (entertainment/self-pleasure) and academic performance.

Lesson 2: A literature review of the effects of social networking sites on secondary school students' academic achievement

H1: Self-efficacy positively predicts students' learning satisfaction during emergency distance learning.

H2: Students' self-efficacy is positively related to their perceived learning outcomes.

H3: Perceived learning outcomes mediate the relationship between self-efficacy and learning satisfaction.

Lesson 3: Task-Technology Fit Model: The Factors Affecting Students' Academic Performance in Higher Education

H1: Social Characteristics (SC) → Task-Technology Fit (TTF)

H2: Technology Characteristics (TEC) → Task-Technology Fit (TTF)

H3: Task-Technology Fit (TTF) → Behavioral Intention to Use (BI)Task-H4: Technology Fit (TTF) → Actual Social Media Use (ASMU)

H5: Behavioral Intention to Use (BI) → Actual Social Media Use (ASMU)

H6: Behavioral Intention to Use (BI) → Student Satisfaction (SS)

H7: Behavioral Intention to Use (BI) → Performance Impact (PI)

H8: Actual Social Media Use (ASMU) → Student Satisfaction (SS)

H9: Actual Social Media Use (ASMU) → Performance Impact (PI)

H10: Student Satisfaction (SS) → Performance Impact (PI)

Lesson 4: Social Network Sites Impact on Learning: Extending the TAM 3 Model to Assess Academic Performance in Higher Education

H1: Subjective Norm (SN) Positively affects Perceived Usefulness (PU)

H2: SN positively affects Image (IMG)

H3: IMG positively affects PU

H4: Job Relevance (REL) positively affects PU

H5: Output Quality (OUT) positively affects PU

H6: Result Demonstrability (RES) positively affects Perceived Usefulness (PU).

H7: Computer Self-Efficacy (SE) positively affects Perceived Ease of Use (PEU)

H8: Computer Anxiety (ANX) negatively affects PEU

H9: Computer Playfulness (PL) positively affects PEU

H10: Perceived Enjoyment (ENJ) positively affects PEU

H11: PEU positively affects PU

H12: SN positively affects Behavioral Intention (BI)

H13: PU positively affects BI

H14: PEU positively affects BI

H15: BI positively affects Use Behavior (USE), and

H16: USE positively affects Academic Performance (AP).

Lesson 5: The Mediating Role of Smart Learning Environment in the Relationship Between Social Media and Academic Performance Among University Students

H1. Social media use positively affects academic performance.

H2. Social media use positively affects the smart learning environment.

H3. Smart learning environment positively affects academic performance.

H4. Smart learning environment mediates the relationship between social media use and academic performance

Lesson 6: The Influence of Social Media on Student Learning Behavior and Its Effects on Academic Achievement

H1: Social media usage has a positive effect on the transformation of students' learning behavior.

H2: The transformation of learning behavior positively affects students' academic achievement.

H3: Social media usage indirectly influences academic achievement through the transformation of learning behavior as a mediating variable.

Lesson 7: Examining Social Media Addiction as a Predictor of Academic Achievement and Academic Procrastination: A Cohort of Undergraduate Students

H1: Self-control has a negative effect on social media addiction and academic procrastination.

H2: Value (dimension of future time perspective) negatively predicts social media addiction.

H3: Distance (dimension of future time perspective) negatively predicts social media addiction.

H4: Commitment (dimension of future time perspective) negatively predicts social media addiction.

H5: Speed (dimension of future time perspective) positively predicts social media addiction.

H6: Value negatively predicts academic procrastination.

H7: Distance negatively predicts academic procrastination.

H8: Commitment negatively predicts academic procrastination.

H9: Speed positively predicts academic procrastination.

H10: Social media addiction positively predicts academic procrastination.

H11: Social media addiction negatively predicts academic achievement.

H12: Academic procrastination negatively predicts academic achievement.

H13: Social media addiction predicts academic achievement through academic procrastination (mediating effect).

Lesson 8: Prediction of social media efects on students' academic performance using Machine Learning Algorithms (MLAs)

H1: The level of exposure to SNSs (e.g., WhatsApp, Facebook, Twitter, Viber, Instagram) significantly affects students' academic performance.

H2: The time duration spent on SNSs has an impact on students' academic performance.

H3: The nature of SNS usage (e.g., academic vs. non-academic) influences students' academic performance.

H4: The use of SNSs in the classroom affects students' academic performance.

H5: The overall rate of SNS usage has a significant relationship with students' academic performance.

Lesson 9: Untangling the Adverse Effect of SNS Stressors on Academic Performance and Its Impact on Students' Social Media Discontinuation Intention: The Moderating Role of Guilt

H1: Social overload has a negative effect on academic performance.

H2: Information overload has a negative effect on academic performance.

H3: System feature overload has a negative effect on academic performance.

H4: Control variables such as gender and age may influence discontinuance intention.

H5: Academic performance has a negative effect on discontinuance intention.

H6: Guilt feelings have a positive effect on discontinuance intention.

Lesson 10: Data Mining Techniques to Analyze the Impact of Social Media on Academic Performance of High School Students

H1: Data cleaning and handling of missing values will improve the quality of input for machine learning models.

H2: Features such as "Technology," "DSM," "WSM," "Romantic," "Living area," and "Desire for higher education" significantly influence students' final grades.

H3: Proper extraction of relevant features will enhance the accuracy of classification models.

H4: Each classification model (Decision Tree, SVM, SGD, Random Forest, Adaboost, Logistic Regression) performs differently in predicting students' final grades.

H5: Certain machine learning models (e.g., Random Forest or Adaboost) will achieve higher accuracy than others.

H6: Using cross-validation techniques provides more reliable results for evaluating model performance.

H7: Accuracy is an appropriate metric to assess model quality in predicting academic performance.

6. Research Objectives

This study delves into the study and evaluation of the impact of social media use on student learning outcomes in today's educational environment. By synthesizing and evaluating information from 10 international studies, the work aims at 5 main objectives.

First, the study aims to determine the relationship between social media use and student learning outcomes. Many studies, including those that show that social media use can have both positive and negative effects on learning outcomes, depending on the way and purpose of use. Factors such as usage time, access frequency, and social media platforms (e.g., Facebook, Instagram, and TikTok) are important factors to consider (Ting Tin Tin et. al. , 2024).

Second, this study focuses on analyzing the mediating and moderating roles of factors such as social media addiction, learning motivation, self-control, and self-efficacy in the relationship between social media use and academic performance. Studies have shown that these psychological-behavioral factors play an important role in explaining why the same type of social media use can lead to different academic outcomes in students (Isaac Kofi Nti et. al. , 2021). The second objective was to examine the mediating and moderating role of psychological-behavioral factors, including social media dependence, learning motivation, self-efficacy, self-control, and mental health status, in the association between social media use and academic achievement. As an illustration, research has shown that social media overuse increases problems such as stress and anxiety in students, resulting in reduced academic performance (Isaac Kofi Nti et. al. , 2021).

The third objective is to clarify the differences in the impact of social media, based on the purpose of use – for academic work or entertainment. Many studies have shown that using social media for learning purposes (group discussions, searching for documents, participating in online learning) can significantly improve learning efficiency as well as student satisfaction (S. A. Barabadi et. al. , 2022). On the contrary, excessive use of social media for entertainment activities often leads to distraction, reduced interest in learning and poor learning outcomes (S. A. Barabadi et. al. , 2022).

The fourth objective is the development and application of theoretical frameworks to examine the impact of social media in educational settings, including constructs such as the TTF Model, the TAM Model, or TAM 3. These models have been used in studies by Barabadi et al.,

showing that when social media meet the requirements of learning, are built in a friendly and easy-to-use way, students tend to exploit them more effectively and achieve better learning outcomes (S. A. Barabadi et. al. , 2022).

The fifth objective, this study aims to present practical solutions to help students use social media platforms effectively in the learning process. Based on the studies of Melese Astatke et al., it can be seen that it is important to establish programs for guidance, awareness raising, time management skills training, and the use of social media for learning purposes. Education administrators should also consider integrating social media platforms into teaching activities, in a controlled and clearly directed manner (Melese Astatke et. al. , 2021).

7. Significance of the Research

Explain why this study is important and who will benefit.

The significance of this study lies in providing a clearer and more comprehensive understanding of how social media affects student learning outcomes, especially in higher education settings. In the context of increasing digital interaction among university students, social media platforms are no longer limited to individual interactions but have become tools for collaborative learning, peer communication, and sharing of academic content. However, existing studies have presented inconsistent views: some emphasize positive impacts on learning outcomes, while others point to risks such as addiction, anxiety, and distraction. These contradictions create an urgent need for further research to clearly identify under what conditions social media effectively supports learning, and when it becomes a hindrance. Based on that, this study is particularly valuable for university students – the main users of social media in the academic environment. It helps students raise awareness of how they use social media and better understand its impacts on their engagement and learning outcomes. The findings from the study not only support students in using social media in a purposeful and effective way, but also help them avoid risks such as abuse or misuse. At the same time, educators and academic institutions can take advantage of these results to develop appropriate interventions, effectively integrate social media into teaching strategies and promote digital competence in training programs. In addition, policy makers and education administrators can also rely on evidence from the study to develop reasonable guidelines and policies on digital behavior, to maximize the benefits and minimize the negative impacts of social media in education. Theoretically, this study also makes significant contributions through the application of the Task-Technology Fit

(TTF) model and the Technology Acceptance Model 3 (TAM 3). Both models are widely used to help students better understand how to apply technology in education effectively and appropriately. Specifically, the TTF (Task-Technology Fit) model emphasizes the importance of the fit between the deployed technology and the characteristics of the learning task. Specifically, when technology is applied properly and meets the specific requirements of the task, learners tend to use that technology more frequently, thereby contributing to improving learning effectiveness (Melese Astatke et. al. , 2021). In addition, the TAM 3 model expands the perspective by focusing on users' psychological cognitive factors, such as perceived usefulness, ease of use, and enjoyment of social media experiences. These factors significantly influence the intention to use social platforms, and increase the likelihood of benefiting from using them if they are properly exploited and serve the right learning goals (S. A. Barabadi et. al. , 2022). This study addresses important research gaps by combining the TTF and TAM 3 models, and broadens the scope of analysis by including important factors such as self-regulation, learning motivation, and engagement – aspects that have not been fully explored in previous studies.

Beyond the academic scope, this study has broader regional and international implications. While many existing studies are limited to specific academic disciplines or demographic groups, the findings of this study can be broadly applied across different educational sectors and contexts, especially in Southeast Asia, where social media use among students is widespread and education systems face similar challenges in effectively implementing social media. As the transition to blended learning in the wake of the pandemic over the past few years has progressed, the insights gained from this study can help educators and institutions optimize the integration of technology to make digital learning more effective and inclusive.

ASSIGNMENT 2

1. Literature Review

This study delves into understanding and assessing the impact of social media use on student learning outcomes in today's educational environment. By synthesizing and evaluating information from 10 international studies, this work aims to achieve five main objectives. The first objective is to determine the relationship between social media use and student learning outcomes.

The first objective of the study is to establish a causal relationship between social media use and student academic performance. Many studies, including articles, have shown that social media use can have both positive and negative effects on academic performance, depending on the way and purpose of use (Ting Tin Tin et. al. , 2024). Factors such as time spent, frequency of access, and social media platform (e.g., Facebook, Instagram, and TikTok) are important factors to consider.

Second, this study focuses on analyzing the mediating and moderating roles of factors such as social media addiction, learning motivation, self-control, and level of confidence in the relationship between social media use and academic achievement. Studies have shown that these behavioral psychological factors play an important role, explaining why the same type of social media use can lead to different academic outcomes in students (Isaac Kofi Nti et. al. , 2021) The second objective was to examine the mediating role and moderating effect of psychological and behavioral factors, including social media dependence, learning motivation, self-efficacy, self-control, and mental health status, in the association between online use and academic performance. As illustrated, research has shown that social media overuse increases problems such as stress and anxiety in students, which in turn leads to lower academic performance

(Ayesha Masood et. al. , 2022). In addition, studies have also emphasized the importance of other personal factors such as current academic performance or self-management (Isaac Kofi Nti et. al. , 2021). The third objective is to clarify the differences in the impact of social media, based on the purpose of use for academic or recreational work. Many studies have shown that using social media for academic purposes (group discussions, searching for documents, participating in online learning) can significantly improve learning efficiency as well as student satisfaction (S. A. Barabadi et. al. , 2022). On the contrary, excessive use of social media for

recreational activities often leads to distraction, reduced learning interest, and poor learning outcomes. The fourth objective is to develop and apply theoretical frameworks to examine the impact of social media in educational settings, including constructs such as the TTF Model, the TAM Model, or TAM 3. These models have been used in studies by Barabadi et al., showing that when social media meet learning requirements, are designed in a friendly and easy-to-use manner, students tend to exploit them more effectively and achieve better learning outcomes (S. A. Barabadi et. al. , 2022).

The fifth objective of this study aims to present practical solutions to help students use social media platforms effectively in their learning process. Based on the studies of Melese Astatke et al., it can be seen that it is important to establish guidance programs, increase awareness, train time management skills, and utilize social media for learning purposes (Melese Astatke et. al. , 2021). Education administrators should also consider integrating social media platforms into teaching activities in a controlled and directed manner.

In recent years, the rapid development of digital technology has significantly changed the way students search for information and engage in learning activities. Among the myriad digital tools, social media has emerged as a phenomenon, not only used for entertainment but also applied to education. Therefore, many studies have investigated the impact of social media use on student learning outcomes. This review synthesizes the results of ten scientific articles, highlighting both important achievements and limitations in existing research. From there, it sets the stage for further research on the complex relationship between social media use and learning outcomes.

Current research and discovery methods

Several studies have used previously established theoretical frameworks to examine the impact of social media on learning. For example, study applied the Task-Technology Fit (TTF) model, which found that when the technological characteristics of social media including accessibility, user interaction, and instant feedback match learning activities such as information exchange, group discussion, and collaborative learning, students tend to utilize these platforms more effectively for learning purposes. Their study of 206 university students confirmed that social and technical factors, along with perceived technology fit, had a significant impact on students' intention to use social media for learning, thereby indirectly improving their learning outcomes (Isaac Kofi Nti et. al. , 2021).

Similarly, Barabadi et al. extended the Technology Acceptance Model 3 (TAM3) by assessing constructs such as perceived usefulness, perceived ease of use, output quality, subjective norms, and perceived enjoyment. Their findings, based on a sample of 400 agricultural students, confirmed that these factors significantly influenced behavioral intentions to use social media, which then influenced actual use and academic achievement. Together, these studies suggest that when students find social media tools both useful and enjoyable, they are more likely to adopt them in an academic context (S. A. Barabadi et. al. , 2022). Further supporting these findings, Kamau found that purposeful, educational social media use positively affected student academic performance, while excessive or recreational use was often correlated with lower academic performance. These studies highlight the importance of distinguishing between different uses and behavioral intentions when assessing the educational impact of social media (Kamau, 2025).

Furthermore, researchers such as Raza et al. have examined the psychological and behavioral mechanisms that link social media use to academic performance. Their studies have introduced mediating variables such as social media addiction, attention span, stress, and self-regulation, suggesting that these internal factors may facilitate or hinder students' ability to benefit academically from their social media engagement (Ayesha Masood et. al. , 2022).

Gaps and limitations in current research

Inconsistency in findings

A major limitation is the inconsistency in conclusions across studies regarding the overall impact of social media on academic performance. For example, found no significant direct relationship between social media use and academic performance, while others reported either positive or negative correlations (Ting Tin Tin et. al. , 2024). Studies have shown mixed results on the relationship between social media and academic performance – some studies show a positive impact, others confirm a negative or no significant relationship due to differences in measurement methods, sample characteristics (country, age, field of study), and theoretical frameworks used (Arago, 2025; Hamidah & Okkita Rizan, 2025; Alison Black at. al., 2025).

Limited geographic and demographic coverage

Most studies focus on a specific country, region, or university (usually Asia or Africa), with a cohort of students from the same major. As a result, the generalizability of the findings is limited. Few studies compare results across multiple cultural or educational contexts. In

addition, there is a lack of attention to the diversity of the student population, including differences in year, field of study, and level of education. Future research should incorporate broader, multicultural samples to increase external validity (Ana Cristina García et. al., 2020;Arago, 2025;Niamat Ullah et. al. , 2025).

Narrow measurement of social media behavior

Another gap lies in how social media use is typically measured. Most studies focus on general metrics such as time spent or frequency of use, without analyzing the specific types of activities students engage in. For example, spending an hour on Facebook in a group discussion is fundamentally different from passively scrolling through TikTok videos. Raza et al. took this a step further by examining short-form video consumption (e.g., film reels), but such analyses remain rare. More detailed investigations of the nature and purpose of social media behavior are essential to understanding its scholarly significance (Hamidah & Okkita Rizan, 2025;Victor Ndubuaku et.al., 2020).

Lack of longitudinal designs or testing

Most existing studies rely on cross-sectional survey designs. There is a lack of longitudinal or experimental studies to test causality between social media use and academic outcomes. This limits the ability to draw causal inferences. Longitudinal or experimental studies are needed to determine how changes in social media behavior over time affect academic outcomes. Furthermore, although many studies examine mediation or moderation effects, few have developed comprehensive, integrative models that incorporate behavioral, psychological, technological, and academic variables into a cohesive framework (Nader Salari et.al., 2025;Guaña-Moya, 2025;Victor Ndubuaku et.al., 2020).

Lack of analysis of mediating and moderating factors

Many studies only examine the direct relationship between social networks and academic achievement, ignoring factors such as addiction, motivation, mental health, digital skills, etc. This leads to the consequence of not clarifying the mechanism of "why" and "under what conditions" social networks are beneficial/harmful (Arago, 2025;Alison Black at. al., 2025;Victor Ndubuaku et.al., 2020).

Lack of multidimensional theoretical integration

Many studies only rely on a single theoretical framework such as TTF or TAM, not integrated with theories of psychology, behavior and digital skills. This leads to the consequence of incomplete analysis of factors related to learning in the digital environment, leading to a gap

in the incomplete integrated theoretical framework, requiring a combination of multiple theories to reflect the modern context (Nader Salari et.al., 2025;Arago, 2025;Alison Black at. al., 2025).

Opportunities for future research

Theoretical Integration

Future research should apply and test integrative theoretical frameworks, incorporating elements from TTF, TAM3, Self-Regulation Theory, Social Cognitive Theory, etc. This would allow for a more comprehensive understanding of how individual, technological, and contextual factors interact.

Diverse Populations and Contexts

Researchers should expand their samples to include students from a variety of disciplines, educational levels, and cultural backgrounds to enhance generalizability and explore context-specific effects.

Detailed Behavioral Analysis

It is essential to distinguish between academic and recreational use, as well as active and passive engagement. Categorizing types of use and linking them to learning goals will yield more precise insights.

Longitudinal and Experimental Methods

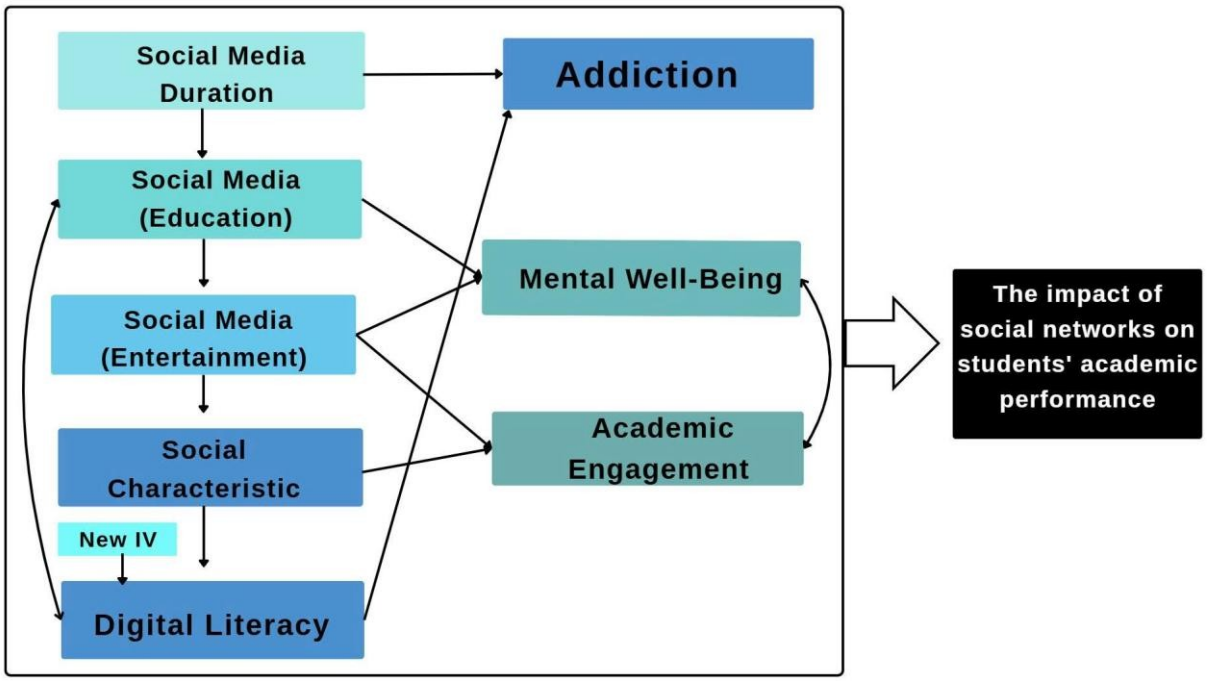
More longitudinal and intervention-based research is needed to establish causal relationships and evaluate the effectiveness of educational strategies that incorporate social media.

Conclusion

Overall, the existing literature provides a valuable foundation for understanding the complex relationship between social media and academic outcomes. However, current findings are fragmented and inconclusive, largely due to methodological limitations, narrow contexts, and inconsistent definitions of key constructs. Future research should aim to bridge these gaps by developing comprehensive models, using diverse samples, and employing robust, time-sensitive methods. In doing so, scholars can better determine when, how, and why social media serves as a beneficial or detrimental influence on student academic success in contemporary higher education.

2. Conceptual Framework

2.1. Conceptual Framework



2.2. Explanation of the research model

This research model is built to explore the impact of social media use on students' learning outcomes, through mediating factors and a new independent variable, Digital Literacy. The model synthesizes important factors from 10 international research articles, while expanding the perspective on digital literacy in the context of modern education.

Social Media Duration is the first independent variable reflecting the overall level of exposure to social media. Excessive use can lead to Addiction, negatively affecting mental health (Mental Well-Being) and the ability to concentrate on studying. Accordingly, Addiction plays the role of a key mediating variable, explaining the indirect impact of social media usage habits on learning outcomes.

In addition, two specific aspects of social media use, Social Media (Education) and Social Media (Entertainment), are also considered. Using social media for learning purposes can help students search for documents, discuss in groups, and develop learning skills through the digital environment. On the contrary, using social media for entertainment purposes has the potential to cause Cyberbullying or reduce interaction with learning, thereby affecting Mental Well-Being and Academic Engagement. These are two key mediating variables that reflect the level of engagement and learning spirit of students. Social characteristics (social characteristics) such as influence from friends, social media culture in the student group, also play an important role. These factors affect Academic Engagement, which is the dependent variable proposed in

this model. The new independent variable, Digital Literacy, represents students' ability to use, evaluate, and create digital information effectively. This is an increasingly important skill in the digital education context, especially when social networks are no longer just entertainment tools but have become formal learning and communication platforms.

Mental Well-Being and Academic Engagement are two central mediating variables in the model, acting as a bridge between factors such as Addiction, Social Media Use, and learning outcomes. When students have good mental health and high levels of learning engagement, they are likely to achieve better learning outcomes.

The impact of social networks on students' academic performance. This is a measure of learning effectiveness through grades, assessment results, and students' feelings about their achievements in the university learning environment.

This model not only clarifies the complex relationship between social networks and learning outcomes, but also extends it in a modern direction by integrating the Digital Literacy factor that reflects students' level of adaptation to the digital learning environment. Understanding this mechanism will help universities and lecturers propose appropriate policies and teaching methods to both take advantage of the benefits of social networks and minimize negative impacts on students' learning.

3. Hypotheses

H1: The amount of time spent on social media is positively related to social media (Education)

Hypothesis H1 suggests that the more time students spend on social media, the more likely they are to use these platforms for educational purposes. The reason for this hypothesis comes from the fact that social media is not only an entertainment tool but also gradually becoming an effective learning support channel, especially in the context of modern education that relies heavily on information technology and digital learning materials.

Nowadays, platforms such as Facebook, YouTube, TikTok, Instagram, Zalo, etc. not only provide entertainment content but also contain a lot of academic content, from lectures, study tips, reference materials to academic discussion groups. When students are exposed to social media a lot, they are more likely to encounter educational content and thus be attracted to using social media in a learning-oriented way. The time spent on social media also means that students have formed the habit of searching, exchanging, and updating information through

these platforms. If they have a positive perception of the role of social media in learning, they will actively follow educational channels, join study groups, or follow lecturers and experts. This relationship is only a general trend and can be influenced by intermediate factors such as learning motivation, self-learning ability, digital skills level, or personal characteristics. Not all students who use social media a lot use it for educational purposes; this also depends on their learning orientation and goals.

H2: Social media (Education) is positively related to addiction

Hypothesis H2 poses an interesting and paradoxical relationship: the use of social media for educational purposes may be positively related to social media addiction. At first glance, this may seem contrary to the conventional wisdom that learning is a positive activity and addiction is a negative consequence. However, when we delve deeper into students' social media behavior, this hypothesis has some merit. One of the main reasons is that the line between using social media for learning and entertainment is becoming increasingly blurred. Many students start using social media to find materials, follow academic pages, watch lecture videos, etc., but in the process, they are easily drawn into content unrelated to learning, leading to uncontrolled use. Over time, this habit can form a dependency, or even an addiction. Learning through social media puts students in constant contact with that platform all day long, increasing the frequency of access and attachment. When used excessively, even if initially for academic purposes, students can still develop symptoms of addiction, such as feeling uncomfortable without social media, losing focus when studying offline, or reducing real-life social interactions. Another factor contributing to this relationship is the design of social media platforms, which are very attractive and "addictive" through personalized algorithms, push notifications, short videos, or feedback mechanisms such as likes, shares, comments, etc. These factors affect users' psychology and make learning via social media unintentionally become a gateway to platform abuse. Therefore, H2 has warned that although using social media for education is positive, if not well controlled in terms of time and usage, it can inadvertently contribute to increasing the risk of social media addiction in students.

H3: Social Media (Entertainment) is positively associated with Mental Health

Hypothesis H3 proposes that the use of social media for entertainment purposes can have a positive impact on students' mental health. This is based on the view that when used appropriately, social media can be a tool to reduce stress, improve mood, and create a sense of social connection, all of which are important factors for mental health. College students often

face academic pressure, family expectations, and social relationships. In this context, entertainment content on social media, such as funny videos, music, online games, creative trends, or viral memes, can help relieve stress, bring laughter, and relax the mind. Studies have also shown that entertainment is one of the natural mechanisms for humans to regulate negative emotions and maintain psychological balance. On the other hand, social media also helps students maintain connections with friends, family, and communities of interest. The feeling of belonging to a social group or being recognized in online interactions contributes to improving self-confidence, reducing loneliness, and anxiety. Especially during periods of social distancing or online learning, social media acts as a "spiritual bridge" to help students maintain social interactions, a key factor in maintaining mental health. This positive impact only occurs when students use social media in moderation and with a positive orientation. If social media is abused or exposed to too much negative content (self-comparison, online bullying, fake news, etc.), the opposite effect, causing anxiety, depression, or decreased self-worth may occur. From here, H3 opens up a research direction to determine the extent and specific conditions under which using social media for entertainment purposes can improve students' mental health.

H4: Social Media (Entertainment) is positively related to Academic Engagement

Hypothesis H4 proposes that students' use of social media for entertainment purposes can promote engagement in academic activities. This relationship may seem indirect, but there are many theoretical and practical bases supporting the assumption that social media entertainment activities can contribute to creating conditions or a positive environment to enhance academic motivation. First of all, entertainment activities are a form of energy regeneration and mental restoration. When students are relaxed through attractive entertainment content such as comedy videos, music, games, funny trends, etc., they can improve their mood, reduce stress, and regain their spirit to continue participating in academic activities. This is especially important in a stressful school environment.

Second, entertainment content on social media is often presented in a creative and lively way, which can stimulate curiosity and creativity in students. Skills such as visual thinking, content creation, quick response, etc., when developed in an entertainment environment, can be "transferred" indirectly to the learning context, especially subjects that require critical thinking, discussion, or presentation. Social media also creates informal learning environments where students are entertained while learning. For example, TikTok pages teach foreign languages through music, Instagram shares study tips, and educational memes help students remember

knowledge faster. When entertainment becomes a potential learning channel, students can be attracted to the learning process in a more natural and active way. Like many other hypotheses in the model, this relationship may be influenced by the duration of use and how students orient their entertainment behavior. If entertainment is overused or uncontrolled, the positive effects may be reversed. H4 is a noteworthy hypothesis to consider whether entertainment is not only a learning tool but also a “mental lever” that indirectly increases learning engagement.

H5: Duration of social media use is positively associated with addiction

Hypothesis H5 states that the longer the duration of social media use, the greater the level of social media addiction. This is a hypothesis that has a solid foundation in behavioral theory, and many previous studies have also confirmed this relationship. In the current context, students often use social media as an indispensable part of their daily lives - from updating news, chatting with friends, entertainment, to studying. When the use becomes too frequent and prolonged without clear limits, users are likely to fall into a state of psychological and behavioral dependence on social media platforms.

These are early signs of addiction. Long usage makes students easily form the habit of checking social media constantly, even feeling restless, impatient, or annoyed if not connected. This behavior can start from reasonable reasons such as studying or keeping in touch, but if not controlled, it will lead to compulsive use, causing negative consequences for academic, social, and mental life. Behavioral psychology also suggests that receiving quick rewards from social networks (such as likes, comments, notifications...) is a factor that stimulates the brain to secrete dopamine, a chemical that causes excitement. When students are exposed to these "social rewards" for too long and too often, they easily become dependent on that feeling and continue to use social networks as an indispensable need - that is, addiction. This further increases the duration of use and shortens the distance to addictive behavior.

H6: Addiction has a negative impact on Mental Health

Hypothesis H6 states that social media addiction has a negative impact on students' mental health. This is one of the important relationships and is of interest to many modern psychological studies, especially in the context of students spending more and more time in the virtual world. First of all, social media addiction is a form of behavioral addiction. That is, users are dependent on an activity (using social media) to the point of losing control and continuing that behavior even though they know it is harmful. For students, when using social media exceeds reasonable limits, they are prone to mental exhaustion, reduced ability to concentrate,

difficulty controlling emotions and increased risk of psychological disorders. One of the common consequences of social media addiction is anxiety and stress. Students often feel the need to check their phones constantly so as not to “miss” new information, leading to FOMO (Fear of Missing Out) syndrome. In addition, constantly comparing yourself to the perfect image on social media can also increase feelings of low self-esteem, dissatisfaction with yourself, and easily lead to mild depression or social anxiety. Social media addiction reduces the time spent on other healthy activities such as rest, sports, real-life communication, or studying. When these aspects are affected, students will feel unbalanced in their lives, leading to sleep disorders, personal disorientation, and increased symptoms of chronic stress. It is important to note that mental health is not just the absence of disease but also includes the ability to cope with pressure, work effectively and contribute positively to society. When social media addiction reduces this ability, it is the negative impact that hypothesis H6 is aiming for. H6 is a practical hypothesis that helps clarify the potential dangers of uncontrolled social media use on students' mental health, a factor that greatly affects academic performance and quality of life.

H7: Addiction has a negative impact on Academic Engagement

Hypothesis H7 suggests that social media addiction has a negative impact on students’ academic engagement. This is a well-founded hypothesis in both theory and practice, as addictive behavior in any form often impairs concentration, self-control, and personal motivation. It also distracts students and reduces their ability to maintain attention for long periods of time, a core element of effective learning. Constantly checking phones, surfing social media in class, or using online entertainment platforms while studying disrupts the learning process. These behaviors directly affect the level of interaction with the lesson, proactively speaking up, discussing, or completing assignments on time, all of which are manifestations of academic engagement. When addicted to social media, students tend to procrastinate. They often surf social networks for “a little entertainment” and then spend hours without completing any learning tasks. This habit, in the long run, reduces learning efficiency and makes students gradually lose motivation to study. Another important factor is the mental exhaustion caused by social network addiction. When dependent on “virtual life”, students easily feel exhausted, lose confidence, are negatively affected by social comparisons, leading to negative moods, a lack of inspiration to study, and reduced interaction with the real learning environment. And it also weakens positive learning relationships such as group study, knowledge exchange with friends, or participation in academic activities. When students are sucked into cyberspace, they tend to

withdraw and have less connection with the surrounding learning environment. H7 is an important hypothesis to emphasize the negative consequences of social network addiction on students' learning process, specifically the decrease in the level of active participation in academic activities.

H8: Mental health positively affects learning engagement

Hypothesis H8 states that when students have good mental health, they will participate in learning more actively. This is an assumption that is reinforced by many studies in the field of educational psychology, showing a close relationship between psychological state and learning behavior. Good mental health is understood as a state in which students feel positive, can control their emotions, effectively cope with pressure, and maintain balance in life. When students have a stable and optimistic spirit, they will have a higher motivation to participate in learning activities such as attending classes regularly, actively speaking up, completing assignments on time, and participating in group study sessions. A positive mental state helps students maintain concentration and better memory. A student who is in a peaceful state, not affected by anxiety, stress, or depression, will easily absorb knowledge and be ready to interact with the learning environment. On the contrary, when the mind is unstable, the brain is easily overloaded and loses the ability to receive information effectively. In addition, good mental health also helps students build positive social relationships in the school environment. They tend to be more sociable, cooperative, and interactive with friends, lecturers, and extracurricular activities, factors that indirectly reinforce active learning participation. A mentally healthy student also often has good self-management skills, knows how to set learning goals, manage time effectively, and maintain discipline in studying. All of these factors contribute to improving the level of learning participation in both attitudes and behaviors.

H9: Mental Health Positively Affects Academic Performance

Hypothesis H9 states that good mental health will contribute to improving students' academic performance. This is one of the foundations in the field of education and psychology because mental health is an important factor affecting the ability to absorb knowledge, maintain motivation, and perform well in learning tasks. Good mental health means that students are able to control stress, maintain a positive mentality, and have stable emotions. These factors help students cope better with the pressure from homework, exams, and academic requirements. When not dominated by anxiety, stress, or depression, students can focus on studying and perform to the best of their ability in tests or learning projects. Good mental health contributes

to maintaining perseverance and the ability to set long-term learning goals. Students with a stable mentality often know how to organize their time, control themselves and are not easily distracted by external factors such as social networks, entertainment temptations, or other worries. This self-management ability helps them learn more effectively and complete their learning tasks fully, thereby improving their learning outcomes. When students have good mental health, they also often have positive relationships with friends, teachers, and the learning environment. They are more likely to integrate, cooperate, and learn from others, which contributes to enhancing the overall learning experience, thereby improving their performance.

In contrast, students with mental health problems such as anxiety, insomnia, and depression often tend to study poorly, miss assignments, lack concentration, and even drop out of school. Emotional barriers can prevent them from being able to develop their full potential, even though they have the same level of education as others.

H10: Learning engagement positively affects Academic Performance

Hypothesis H10 asserts that the level of student engagement in learning has a positive impact on their academic performance. This is one of the clear relationships and has been proven by many educational theories as well as school practices. When students actively participate in the learning process, they will have the opportunity to access deeper knowledge, understand lessons better, and improve their learning outcomes in a sustainable way. Learning engagement is demonstrated through many aspects, including attending classes regularly, speaking constructively about lessons, participating in group discussions, completing assignments on time, asking questions when they do not understand lessons, participating in academic activities outside the classroom, etc. These behaviors reflect students' initiative and commitment to learning. When students are so active, they not only acquire knowledge but also understand how to apply it, thereby improving their thinking ability and learning outcomes. Participating in learning helps students develop important academic skills such as critical thinking, teamwork, time management, and problem solving. These skills contribute to the ability to solve exercises, take tests, and carry out projects more effectively, which is a decisive factor in academic achievement in the higher education environment. Being encouraged, supported, and recognized also boosts learning motivation, helps them to be more confident, and try harder in their studies. Students who are less involved in learning often only learn passively, do not respond to lectures, easily fall behind in knowledge, and have difficulty achieving good results. They may miss

important content, do not understand the requirements of the subject, and lack the necessary skills to handle exams and assignments.

H11: Social characteristics positively influence learning engagement

Hypothesis H11 suggests that students' social characteristics can positively influence their level of learning engagement. "Social characteristics" here include factors such as family background, income, living environment, social support, learning culture, and level of engagement with the academic community. These factors play a key role in shaping the motivation, behavior, and learning attitudes of students from families with a stable social background, such as parents with high education, sufficient income, or positive attitudes towards learning are often inspired and have higher learning expectations. Support from the family helps students feel encouraged, secure, and willing to invest time and effort in learning activities, thereby increasing their level of engagement. Students who are socially integrated, have good communication skills, and regularly interact with classmates, lecturers, or participate in study groups, academic clubs, also tend to participate more actively in learning. These social connections help students feel like they are part of the academic community, increase intrinsic motivation, and attachment to the school. Living environment is also an part of important social characteristics. Students living in urban areas, with access to the internet, libraries, and quality learning spaces, will have an advantage in participating in learning. On the contrary, students who lack social conditions may be limited in resources, learning opportunities, and even learning confidence.

H12: Social characteristics positively affect Digital Skills

Hypothesis H12 states that students' social characteristics have a positive influence on their level of digital skills development. Digital skills here are understood as the ability to proficiently use digital tools, information technology, online learning platforms and social networks effectively, serving the goals of learning and self-development. Students from families with high economic conditions (for example, stable income, parents working in professional fields, with good education) often have the opportunity to access technology earlier. They have the conditions to own technology devices (computers, smartphones, high-speed internet), and are instructed on how to use technology, which creates a favorable foundation for their digital skills to develop quickly and comprehensively. Students in urban areas often have the conditions to be exposed to new technology, have many opportunities to practice digital skills in learning and life, as well as access online courses, digital libraries, and learning forums. Meanwhile,

students in rural or remote areas may be limited in terms of technology infrastructure, affecting their ability to practice digital skills. Students who are socially active, regularly participate in extracurricular activities, student projects, or digital startups, etc., also have many opportunities to gain practical experience with digital tools, thereby enhancing digital skills through practice. Building a social environment that supports technology will be a solid foundation for students to develop comprehensive digital skills.

H13: Social Media (Education) Positively Affects Digital Skills

Hypothesis H13 states that the use of social media for educational purposes can contribute to improving students' digital skills. This is a very reasonable relationship in the context of strong digital transformation in education and modern society, when learning is no longer limited to books, but expanded to the online environment, where students are required to be proficient in using technology. When students use social media for learning purposes, they are forced to interact with many digital tools and platforms such as Google Meet, Zoom, Facebook Groups, YouTube, Zalo, Google Drive, or academic forums. This process helps them improve their ability to search for information, store documents, process text, share content, and work in online groups, which are core competencies in digital skills. Accessing learning content through social media also helps students become familiar with different digital formats such as videos, infographics, podcasts, slideshows, PDF documents, and Google Forms. As students understand how to use and evaluate information from these formats, they are gradually developing digital thinking skills, information processing capacity, and new media skills. Another benefit is that students will learn how to protect personal information, manage digital identities, and behave culturally in cyberspace, which are increasingly important aspects of a comprehensive definition of digital skills.

H14: Digital skills positively affect learning engagement.

Hypothesis H14 asserts that when students have good digital skills, they will participate more actively in learning. In the context of modern education, where technology and online learning methods are increasingly popular, digital skills are not only an advantage but also a necessary condition for students to learn effectively. Digital skills help students easily access online learning resources such as e-documents, video lectures, MOOC courses, digital libraries, and online group learning tools. With good digital skills, students are no longer dependent on traditional learning methods but can actively seek knowledge and study anytime, anywhere. This increases their initiative and flexibility in the learning process, leading to higher levels of

learning engagement. It also helps students interact effectively with lecturers and classmates through digital learning platforms such as Google Classroom, Moodle, Microsoft Teams, Zoom, or academic social networks. When students are able to use these tools, they can discuss in groups, post questions, share documents, participate in online learning, or submit assignments on time, all of which are concrete manifestations of active learning engagement. When students have digital skills, they will feel more confident in the modern learning environment. This confidence leads to a positive attitude towards learning, motivating them not to hesitate to ask questions, participate in discussions, or try new forms of learning.

H15: Digital skills positively influence learning performance.

Hypothesis H15 states that students' digital skills have a positive impact on learning performance, that is, the actual learning outcomes that students achieve. In the era of digital education, this is a well-founded hypothesis, as digital skills have now become an indispensable part of the learning process at the university level. Digital skills help students effectively use online learning tools such as learning management systems (LMS), word processing software, spreadsheets, presentations, academic search engines (Google Scholar, ResearchGate), as well as language learning software, simulation tools, and AI learning support platforms. By mastering these tools, students can improve their learning productivity, complete assignments better, and achieve higher grades. Digital skills help students shorten the gap with global knowledge. Knowing how to search, evaluate, and apply reliable online resources not only helps students update their knowledge quickly but also develops a solid academic mindset. This contributes significantly to improving the quality of assignments, reports, presentations, and the ability to apply knowledge in practice, thereby improving learning performance. Equally important, digital skills also help students present their learning outcomes in a more professional and convincing manner, from writing formats, designing presentation slides, creating academic videos, to analyzing data with specialized software. These are all factors that are highly appreciated by lecturers and contribute to final learning outcomes.

4. Research Methodology

4.1 Research Design

This study uses a quantitative research method with a cross-sectional survey design to examine the relationship between social media use, digital literacy, learning engagement, mental health, and academic performance of university students. The research model is built on

previous empirical studies, and expanded by introducing a new variable, digital literacy, which reflects learning ability in today's digital environment.

The quantitative method was chosen because it allows for the collection of measurable data that can be statistically analyzed from a large number of participants. This method helps test hypotheses using standardized analysis tools, while controlling for multiple variables at the same time and easily generalizing the results to the target population. The cross-sectional design means that data will be collected at a single point in time. Although it does not allow for testing causality over time like a longitudinal study, this design still allows for identifying relationships and influences between variables, which is particularly effective in studying digital behavior. Independent variables include the duration of social media use, the purpose of using social media for learning and entertainment. Intermediate variables such as social media addiction, mental health, and learning engagement help explain the indirect influence mechanism on learning outcomes. The new variable digital skills acts as both an intermediate outcome and a predictor of learning outcomes.

All variables were measured on a 5-point Likert scale (1 = Strongly disagree to 5 = Strongly agree), which makes it easy to handle statistics and ensures consistency across sections. Despite its clear advantages, the study is aware of some limitations such as self-reported data can lead to cognitive bias, and cross-sectional data does not reflect changes over time. However, with a clear design, systematized model and valid measurement tools, the study is expected to provide insight into the impact of social networks on learning outcomes through intermediary mechanisms.

4.2 Sampling techniques

The research subjects are full-time university students studying at universities in Vietnam - a group classified as Generation Z, growing up with technology and having a very high level of social network usage in both study and personal life. The study uses a non-probability sampling method, combining two popular techniques: convenience sampling and snowball sampling. Convenience sampling with easily accessible students (in classes, clubs, Zalo /Facebook groups) will be surveyed first.

Snowball sampling is that the first respondents will be invited to introduce friends and acquaintances at school who also use social networks for at least 1 hour per day to participate

in the survey. This method helps to expand the sample to other faculties and schools, increasing diversity.

Sampling criteria

- Full-time university students
- Use social networks at least 1 hour per day
- Agree to participate in the survey voluntarily and fully answer the questionnaire.

The desired sample size is from 300 students or more, each person will take 10-15 minutes, ensuring enough for multivariate regression analysis and statistical significance in testing the theoretical model.

4.3. Data Collection Methods

Data will be collected through an online self-administered questionnaire using Google Form or the school's support platform. The questionnaire is designed with 6 main parts, corresponding to the variables in the research model

- Social media use: measuring average time/day and purpose of use (education/entertainment)
- Social media addiction: using the validated SMAS scale
- Mental health: measured by the WHO-5 index;
- Level of learning participation: using the UWES-S scale (student version)
- Digital literacy: measured according to the DigComp framework (EU) including information evaluation, content creation, and digital communication skills

Self-reported learning outcomes GPA and satisfaction with learning achievements. All items are on a 5-point scale, except for GPA, which can be open-ended or have a score range.

- 1 = Strongly disagree
- 2 = Disagree
- 3 = No opinion
- 4 = Agree
- 5 = Strongly agree

The data will be automatically encoded, securely stored, and used for research purposes only. Before the official survey, the questionnaire will be tested with 20–30 students to ensure clarity and context appropriateness.

4.4. Statistical Analysis Methods

After collecting data, the analysis will be carried out through 5 main steps

Step 1: Descriptive analysis & data cleaning

Check for missing data, remove invalid responses. Calculate the mean, standard deviation, frequency for variables and demographic characteristics.

Step 2: Reliability and validity test

Calculate Cronbach's Alpha for each scale ($\alpha \geq 0.70$ is acceptable).

Evaluate EFA to check factor groups, and if necessary, conduct CFA using AMOS to confirm the structure.

Step 3: Test regression assumptions

Check linearity, normal distribution, homoscedasticity and multicollinearity (VIF < 10) to ensure the regression model is valid.

Step 4: Multiple linear regression analysis

Use multivariate regression to test the relationship between independent variables (social media use, addiction, digital literacy...) and the main dependent variable (academic performance) ($p < 0.05$).

Step 5: Mediation analysis

Use the PROCESS Macro – Model 4 tool to test the indirect effect (e.g. Addiction → Well-being → Academic Performance). Use bootstrapping of 5000 replicate samples to test the

List all cited references in APA format.

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